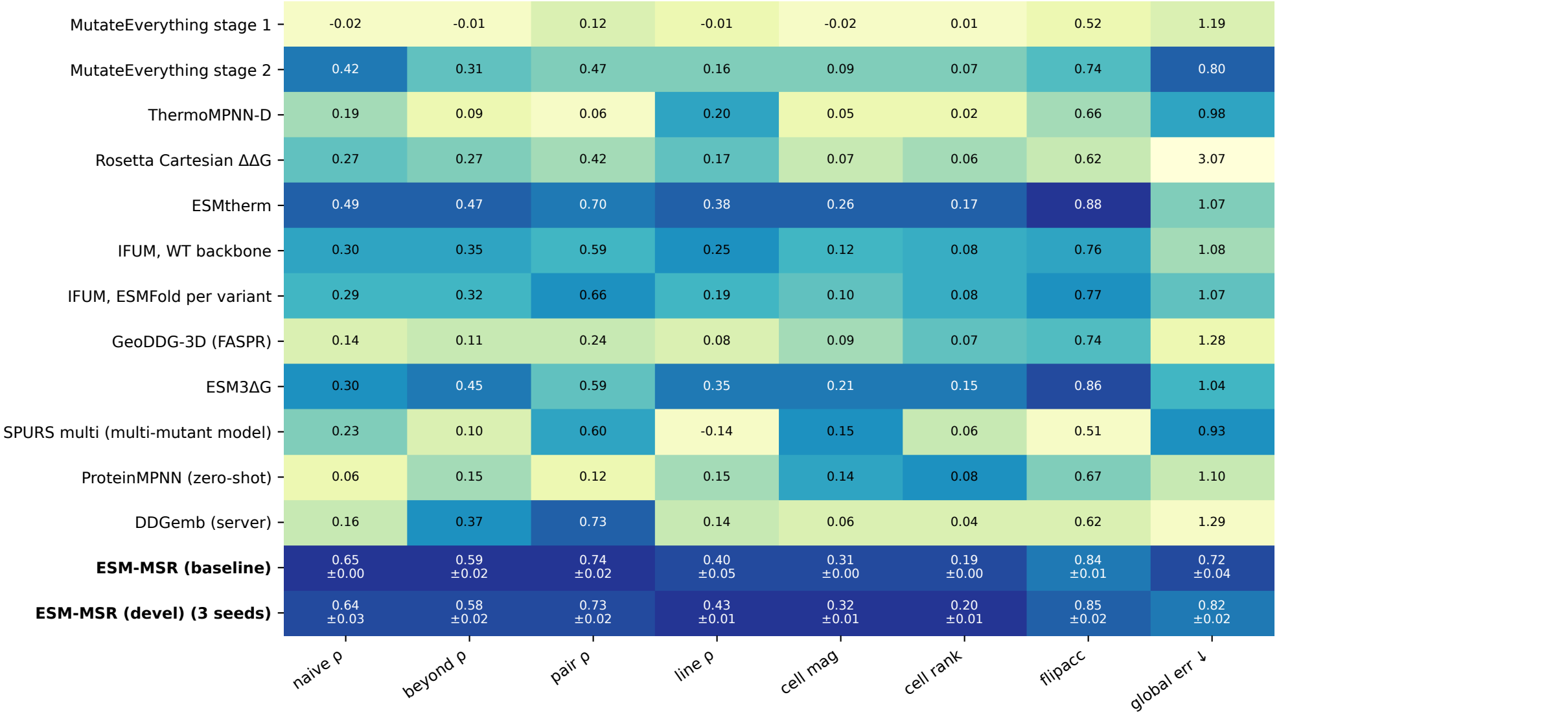
Validation (8,494 doubles): the epistasis hierarchy (table 2.1); colour = rank within the column, darker = better; ± = seed sd
MutateEverything stage 1
MutateEverything stage 2
ThermoMPNN-D
Rosetta Cartesian ΔΔG
ESMtherm
IFUM, WT backbone
IFUM, ESMFold per variant
GeoDDG-3D (FASPR)
ESM3ΔG
SPURS multi (multi-mutant model)
ProteinMPNN (zero-shot)
DDGemb (server)
ESM-MSR (baseline)
ESM-MSR (devel) (3 seeds)
-0.02 -0.01 0.12 -0.01 -0.02 0.01 0.52 1.19
0.42 0.31 0.47 0.16 0.09 0.07 0.74 0.80
0.19 0.09 0.06 0.20 0.05 0.02 0.66 0.98
0.27 0.27 0.42 0.17 0.07 0.06 0.62 3.07
0.49 0.47 0.70 0.38 0.26 0.17 0.88 1.07
0.30 0.35 0.59 0.25 0.12 0.08 0.76 1.08
0.29 0.32 0.66 0.19 0.10 0.08 0.77 1.07
0.14 0.11 0.24 0.08 0.09 0.07 0.74 1.28
0.30 0.45 0.59 0.35 0.21 0.15 0.86 1.04
0.23 0.10 0.60 -0.14 0.15 0.06 0.51 0.93
0.06 0.15 0.12 0.15 0.14 0.08 0.67 1.10
0.16 0.37 0.73 0.14 0.06 0.04 0.62 1.29
0.65 ±0.00 0.59 ±0.02 0.74 ±0.02 0.40 ±0.05 0.31 ±0.00 0.19 ±0.00 0.84 ±0.01 0.72 ±0.04
0.64 ±0.03 0.58 ±0.02 0.73 ±0.02 0.43 ±0.01 0.32 ±0.01 0.20 ±0.01 0.85 ±0.02 0.82 ±0.02
naive ρ
beyond ρ
pair ρ
line ρ
cell mag
cell rank
flipacc
global err ↓